TED(10)-4066 Reg .No…………………………………..

(Revision-2010) Signature:……………………………..

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING /TECHNOLOGY

MODEL QUESTION PAPER FOR COMPUTER COMMUNICATION FUNDEMENTALS

(MEDICAL ELECTRONICS)

[TIME : 3 hours]

(Maximum Marks : 100)

PART – A

(Answer all questions in one or two sentences. Each question carries 2 marks)

I

1. Vestigial sideband system is used in television transmission. Why?
2. In a PCM system having M levels and each level having magnitude N, What is the maximum quantization error?
3. In Hamming code if there are 3 parity bits used, what is the maximum possible message bits?
4. What is the highest audio modulating frequency used in commercial FM broadcasting?
5. Which are the store and forward switching schemes used in data communication?

(5x2=10)

PART –B

(Answer any five questions. Each question carries 6 marks)

II

1. With necessary diagrams differentiate between bit rate and baud rate.
2. Compare the features of OFC and Coaxial cable.
3. Calculate the modulation index of AM.
4. Differentiate between asynchronous and synchronous transmission.
5. List the functions of Application layer in ISO OSI reference model.
6. Calculate the percentage power saving in single side band suppressed carrier AM transmission with
   a) 100 % modulation b) 75 % modulation
7. With diagram list the various network topologies.

(6x5=30)

PART –C

(Answer one full question from each unit. Each question carries 15 marks)

III

1. Explain the various broadcast bands used in communication. 8
2. With circuit diagram, explain the operation of collector modulator. 7

OR

IV

1. Differentiate between low level and high level modulation. 8
2. Explain AM demodulation technique with necessary diagrams. 7

Unit – II

V	1. With block diagram explain digital communication system.	8
	2. Explain how channel bandwidth limits the data transmission rate in digital communication.	7

OR

VI	1. With necessary diagrams explain Time division multiplexing.	8
	2. State Sampling theorem. What is its significance?	7

Unit – III

VII	1.Explain different types of ARQ systems in error handling.	10
	2. Differentiate between serial and parallel transmission.	5

OR

VIII	1. What is MODEM? Why is it essential in digital data communication?	6
	2. Briefly describe the following a) ASK b)PSK c)FSK	9

Unit – IV

IX	1. Briefly explain the concept of ISDN.	7
	2. Explain the most widely used switching technique in data communication.	8

OR

X	1.Explain the necessity of using protocols in data communication.	7
	2. Write short notes on a) Bridges b) Hubs	8